[INCH-POUND]
A-A-1974A
October 27, 1995
(SUPERSEDING)
A-A-1974
May 19, 1982

COMMERCIAL ITEM DESCRIPTION

FLASHLIGHT

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (cid) covers tubular flashlights for general and industrial uses. These flashlights are not intended for environments containing explosive gases or extreme temperatures.

2. CLASSIFICATION

2.1 Classification. The flashlights shall be of the following types, classes, and styles (see 7.3):

Type I - Standard service
Type II - Rough service

Class A - 2-cell
Class B - 3-cell

Style 1 - Without holding magnet
Style 2 - With holding magnet

3. SALIENT CHARACTERISTICS

3.1 General description. The flashlights shall be of tubular design consisting of a battery housing, reflector, lens, lamp, switch, and additional components as required. The flashlights shall be designed to operate on type "D" (ANSI/NEDA No. 13A or 13D) dry cell batteries in accordance with ANSI C18.1. Batteries shall not be furnished.

3.2 Design and construction.

3.2.1 Types.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: General Services Administration, Engineering Group (7FXEE), 819 Taylor St., Fort Worth, TX 76102

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3.2.1.1 Type I. Type I flashlights shall be a commercial type designed for general usage and shall meet all requirements of this cid. The case, caps, and lens rings may be metal, plastic, or combination thereof. Metal case flashlights shall be either steel or aluminum, with a chrome plated or painted finish. The color of the case shall be the manufacturer's option with the exception of olive drab.

3.2.1.2 Type II. Type II flashlights shall be watertight and designed for rough service usage. Type II flashlights shall meet all applicable requirements of this cid. The case, caps, lens rings, and storage compartments for spare lens shall be of molded plastic such as polypropylene, polycarbonate, or other suitable material. Switch contacts shall be of brass, bronze, copper coated steel, or other suitable material. Gaskets shall be provided at any joint through which moisture could enter the interior of the flashlight. The gaskets shall be resistant to oil and water. Unless otherwise specified, the battery case color shall be olive drab to approximately match color number 14087 of FED-STD-595. Colors of the end cap and lens ring shall be the manufacturer's option.

3.2.2 Classes.

3.2.2.1 Class A. Class A flashlights shall hold two "D" cell batteries and use a PR2 lamp.

3.2.2.2 Class B. Class B flashlights shall hold three "D" cell batteries and use a PR3 lamp.

3.2.3 Lens. The lens shall be formed from clear plastic that will not discolor or be affected by salt water, oil, or gases. Glass lens may be used in type I flashlights provided all other requirements of the cid are fulfilled. The light transmission of the lens shall be a minimum of 85 percent. The diameter of the lens shall be a minimum of 1-1/2 inches (3.8 cm).

3.2.4 Reflector. The contour of the reflector shall be essentially parabolic in design. The reflector shall be designed and finished in a manner which assures a coefficient of reflection meeting the requirements of this cid. The reflector shall have sufficient rigidity to prevent distortion when the lamp and batteries are in place. The reflecting surface or facets shall be smooth and free from bubbles or pits.

3.2.5 Battery contact spring. The length and design of the rear contact spring shall accommodate the required number of batteries. The spring shall compress a minimum of 3/8 inch (9.5 mm) and exert a force of not less than 2 pounds (8.9 N).

3.2.6 Roll. The flashlight fully assembled, with batteries, shall be constructed so that when placed on a hard stationary flat surface, inclined 5 degrees for type I and 10 degrees for type II, the flashlight shall not slip, slide, or roll more than 4-1/2 inches (11.4 cm).

3.2.7 Suspension holder (type II only). The type II flashlight shall be equipped with a suspension holder capable of supporting without distortion a weight of 15 pounds (6.8 kg) for two cell flashlights and 25 pounds (11.4 kg) for three cell flashlights. The loop of the holder

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shall be designed to seat firmly into the end cap or lay flat against the battery case.

3.2.8 Lamp and lampholders. The lamps furnished with the flashlights shall be miniature flange base, tungsten filament, single contact, PR2 or PR3. When specified, the lamps provided shall be halogen or krypton bulbs such as KPR-102 or KPR-103, or similar. The lamps shall be solidly retained in place in the lampholder. When specified, the flashlights shall have a secure storage compartment for an extra lamp provided by the manufacturer. The storage compartment shall be designed to be easily accessible and hold the spare lamp securely.

3.2.9 Switch. The switch shall be mounted on the side of the case and shall be solidly retained in any position ("off", "on", or "flash"). Switches shall resist accidental movement to the "on" position. The switch shall be designed to permit easy operation of one finger of the hand holding the flashlight. The switch shall remain operative after 10,000 cycles from "off" to "on" and back to "off" when operated at 15 to 20 cycles per minute. The switch cycle test shall be performed with a constant DC voltage of 2.9 to 3.1 volts across the switch and a total closed-circuit resistance of 0.9 to 1.1 ohm. Switches on type II flashlights shall include a "flash" position.

3.2.10 Threads. Matching threads on the caps, rings, and case shall be capable of being tightened using a torque of 25 inch-pounds maximum. Threads of the type I flashlights shall not strip or ride over those of mating parts when a minimum torque of 50 inch-pounds is applied. Threads of the type II flashlights shall not strip or ride over those of mating parts when a minimum torque of 70 inch-pounds is applied.

3.2.11 Case. Type I flashlights may be either one-piece body construction or made with threaded end caps. Type II flashlights shall be made with end caps. The design of the case shall preclude metal parts of the flashlight from creating a closed circuit with the batteries, except with the switch in the "on" or "flash" position. The design of the case shall also preclude internal parts from abrading or otherwise damaging the batteries.

3.2.11.1 Styles. Style B flashlights shall be furnished with a permanent magnet attached to the case. The magnet shall be capable of holding the flashlight (with alkaline batteries included) to a vertical flat unpainted steel plate or sheet 0.060" or thicker for a minimum of 16 hours without movement. Style A flashlights shall be furnished without a magnet.

3.3 Performance.

3.3.1 Illumination. The beam candlepower (candelas) of the concentrated beam shall meet or exceed the values given in table I when the beam is projected onto a screen having 4 and 8 inch circles (10.2 and 20.4 cm) at a distance of 10 feet (3 m). The test voltage shall be 1.2 volts per cell. The photometer shall be calibrated using a reference standard lamp with a color temperature of 2,856 degrees Kelvin. The photoreceptor shall have a minimum diameter of 1 inch (2.5 cm) and be equipped with a light filter to yield a response curve matching that of the International Commission on Illumination Standard Observer Curve (IES Handbook and IES LM 20). The readings shall be

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taken at the center and on concentric 4 inch (10.2 cm) and 8 inch (20.3 cm) circles. The calculation of beam candlepower shall be as follows:

$$I = E \times D^2$$

Where: I = Luminous intensity of the light beam, candelas
E = Illuminance reading of photometer, foot-candles
D = Distance between lens of the flashlight and photoreceptor, feet

Type	Illumination Candle Power			
Class Style	Center and points on 4 inch circle		8 inch circle, 4 points	
	Minimum	Minimum average	Minimum	Minimum average
I - A - 1	540	720	270	315
II - A - 2	540	720	270	315
I - B - 1	1080	1350	450	500
II - B - 2	1080	1350	450	500

3.3.2 Durability. Type I flashlights with batteries installed shall show no evidence of damage when dropped from a minimum height of 3 feet (0.91 m) on to a concrete surface for a minimum of 6 drops. Type II flashlights (with batteries installed) shall show no evidence of damage when dropped from a minimum height of 5 feet (1.5 m) on to a concrete surface for a minimum of 6 drops. The flashlights shall be completely operable after this test; however, lamp filament damage shall not be a cause for rejection.

3.3.3 Water resistance, type II. Type II flashlights shall be resistant to salt water entry. The interior of the flashlight shall remain dry after immersion in a salt water bath for 1 hour at a depth of 3 feet (0.9 m). The salt water shall have a minimum specific gravity of 1.04 and shall be at a temperature of 72 to 82 degrees Fahrenheit.

3.4 Workmanship. The flashlights shall be free from defects such as sharp edges, improper plating, painting, discoloration, chipping, cracking of the finish, or other visual defects which detract from the appearance or impair the serviceability. The flashlights shall function without any assembly required with the exception of installing batteries.

3.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Part Identification Number (PIN). The following part identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

Example for CID A-A-1974:

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AA1974-IA1      (Type I, Class A, Style 1)
   |   |||
   |   |||------ Style (1 or 2)
   |   ||------ Class (A or B)
   |   |----- Type (I or II)
   | ----- Commercial Item Description Number
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7.2 Sources of documents.

7.2.1 Federal standards are available from GSA-FSS Specification Section (3FB-W), Suite 8100, 470 E. L'Enfant Plaza, S.W., Washington, DC 20407.

7.2.2 ANSI standards are available from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.2.3 Illuminating Engineering Society handbooks and standards are available from the Illuminating Engineering Society of North America, 120 Wall Street, 17th Floor, New York, NY 10005.

7.3 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable).

a. Title, number, and date of this commercial item description.
b. Type, class, style, and subtype required (see 2.1).
c. If halogen or krypton lamps are required.
c. If the flashlight must include a spare bulb.
c. Preservation, packing, and marking required.

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7.4 National Stock Numbers (NSN's). The following NSN's correspond to this commercial item description.

NSN	Type	Class	Style	Subtype
6230-00-163-1856	II	A	2	A
6230-00-164-6605	II	A	1	A
6230-00-242-0005	I	A	1	A
6230-00-242-0008	I	B	1	A
6230-00-643-3486	I	B	1	A
6230-00-781-3671	I	A	1	B

MILITARY INTERESTS:

NONE: DoD has no registered interest in this Commercial Item Description until further notice.

PREPARING ACTIVITY:
GSA - 7FXEE

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